bash scripting – arguments and variables

Adapted from materials by Dr. Carrier



Wikimedia

Jealousy

Think about other commands:

```
echo “Hello world!”
```

```
cat file.txt
```

Wouldn’t it be nice if we could take in arguments too?

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- `$0` - command used to call our program
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- Numbers > 9 need braces: `${10}`

Arguments - Special cases

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Access all arguments

- As an array: $@
- As a string: $*

Variables

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Assignment:

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filename=foo.txt
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Accessing: Again use `$name`:

```
echo “I am $age years old!”
```

Variables

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Combine with variables:

```
num_lines=$(grep “dog” file.txt | wc -l)
```

A note from ol’ Dr. Ferg

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```
$filename
```

Or like this:

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Generally, I *always* use `${var}` for my variables to avoid any confusion.

Quotes

Double quotes: expand variables inside

Single quotes: Take everything as a literal

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Common operators:

+, -, *, /, %, ++, --, **

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NOTE: To work with fractional numbers, use another command. Example:

```
echo "scale=2; 5.0 / 2.0" | bc
```

Reading use input interactively

I typically prefer command-line arguments However, you can wait for the user to type a line in the terminal with:

```
read var
```

And then you can access it with `$var`, `var` being whatever variable name you'd like